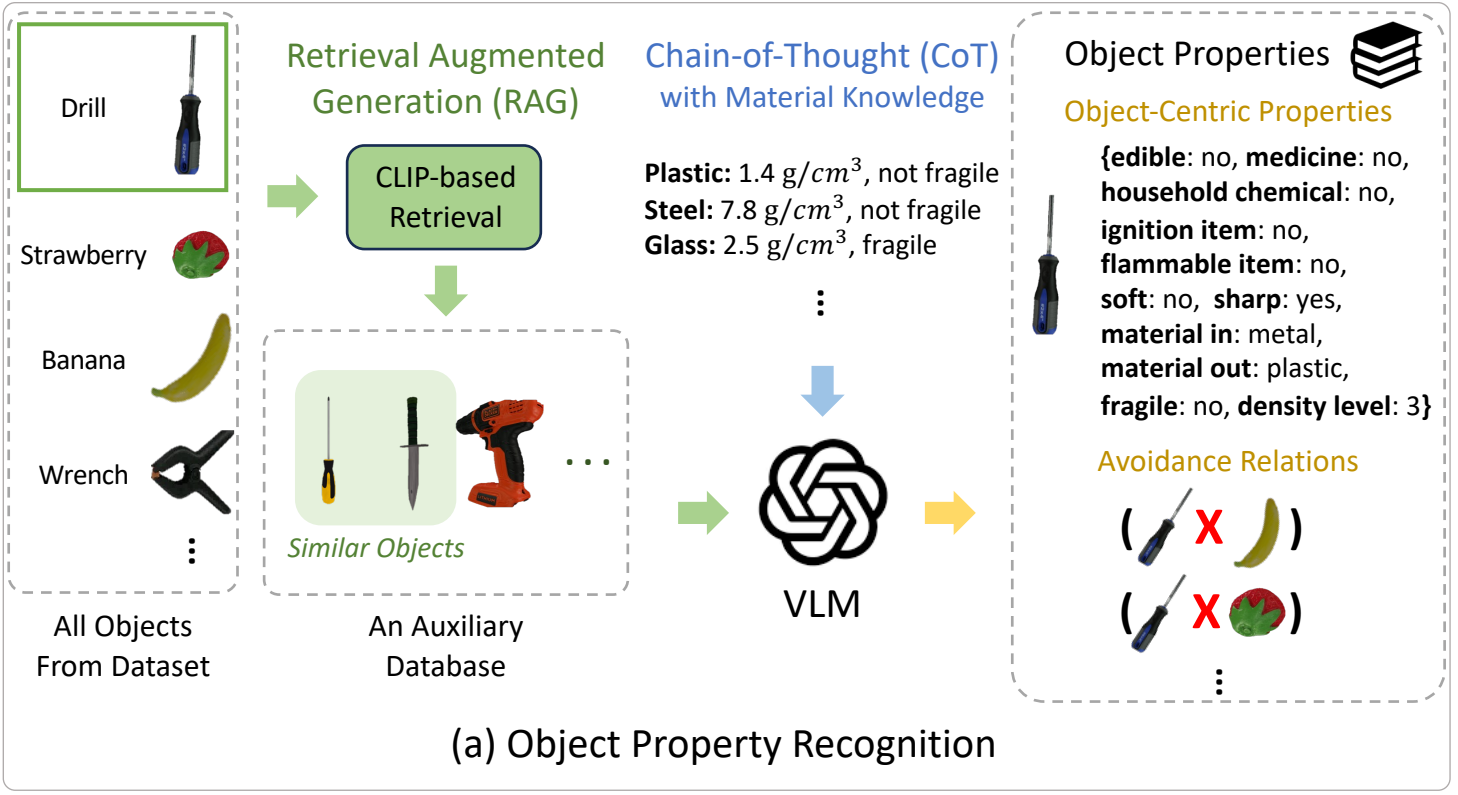
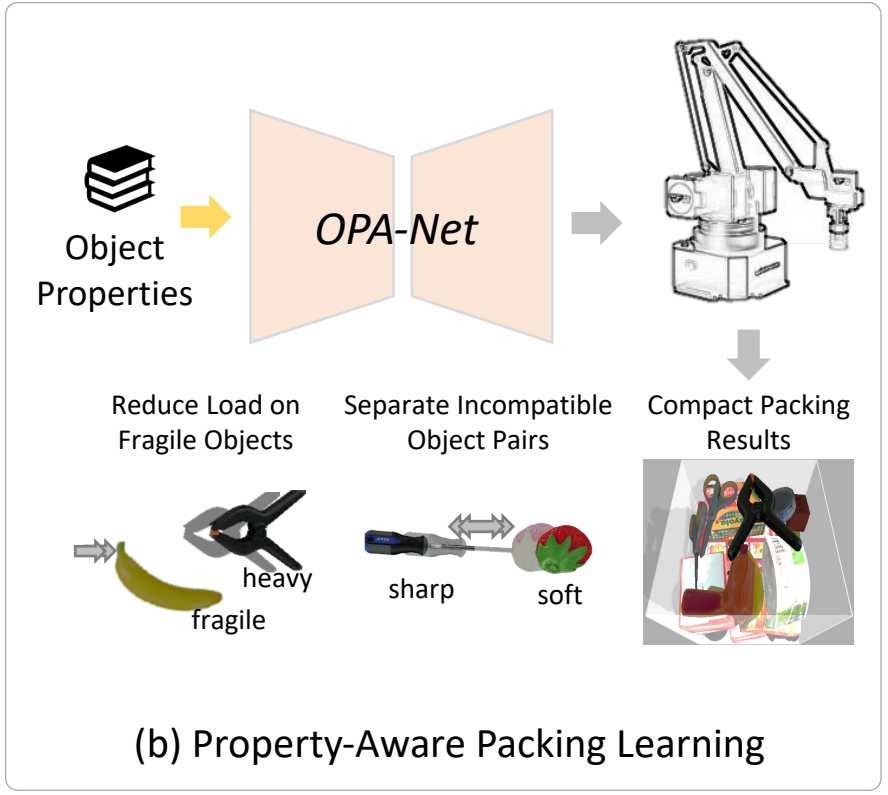

Drill
Strawberry
Banana
Wrench
All Objects From Dataset
Retrieval Augmented Generation (RAG)
CLIP-based Retrieval
Similar Objects
An Auxiliary Database
Chain-of-Thought (CoT) with Material Knowledge
Plastic: 1.4 g/cm³, not fragile
Steel: 7.8 g/cm³, not fragile
Glass: 2.5 g/cm³, fragile
VLM
Object Properties
Object-Centric Properties
{edible: no, medicine: no, household chemical: no, ignition item: no, flammable item: no, soft: no, sharp: yes, material in: metal, material out: plastic, fragile: no, density level: 3}
Avoidance Relations
(a) Object Property Recognition


Object Properties
OPA-Net
Reduce Load on Fragile Objects
heavy
fragile
Separate Incompatible Object Pairs
sharp
soft
Compact Packing Results
(b) Property-Aware Packing Learning